



OTA Updates with Real Time Diagnostics via CAN Bus

[GitHub Project Link](#)



Project Information

University:	Beni-Suef University
Faculty:	Faculty of Engineering
Major:	Communication and Electronics Engineering
Supervisor:	Dr. Mohamed Mourad

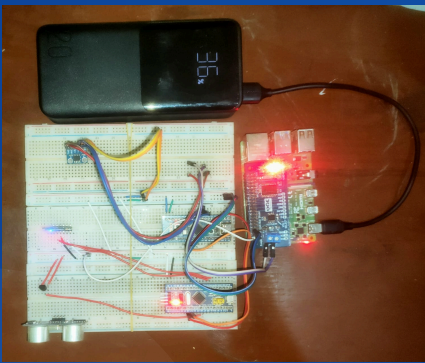
Team Members

- 1. Ahmed Ragab Mogoda
- 2. Amar Ragab Youssef
- 3. Mahmoud El Dinary Mohamed

Project Objectives

This system seamlessly integrates **Real-Time vehicle diagnostics** with robust **Over-The-Air firmware updates** all carried over the CAN bus. At its heart lie two **STM32F103 microcontrollers** running FreeRTOS: one (App ECU) gathers sensor data (LM35 temperature, ultrasonic distance), controls motors, and exposes UDS diagnostic services (session control, security access, read/clear DTC); the Main ECU's bootloader manages signed firmware chunks erasing, programming, and safely switching between application slots.

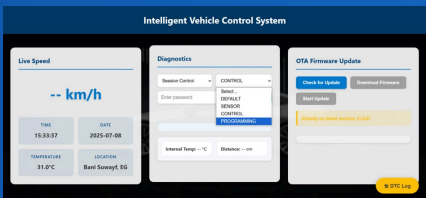
A **Raspberry Pi 4** acts as the user gateway: a **SocketCAN + MCP2515** interface streams live speed frames, while a Flask server and SSE-powered web dashboard display telemetry (speed, temperature, distance), diagnostic trouble codes, and OTA progress. On-demand diagnostics and firmware updates are handled by dedicated UDS and OTA clients, with firmware hosted on a cloud REST API (e.g., AWS S3).



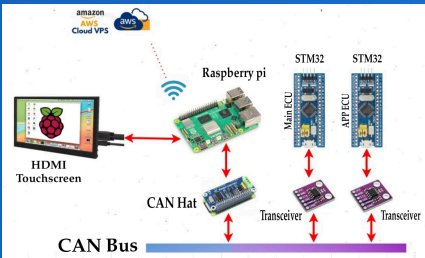
Hardware Implementation



System Integration



Web Dashboard



System Overview